

YERMAN J. MEREL G.

Field Service Mechanical Engineer | Heavy Industrial Equipment | International Projects

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PROFESSIONAL SUMMARY

Mechanical Field Engineer with international experience supporting heavy industrial mechanical systems across large-scale infrastructure and tunnelling projects. Experienced in installation, operation, maintenance, troubleshooting, and validation of complex mechanical equipment, including hydraulic systems, piping systems, and rotating machinery components under demanding field conditions. Proven ability to execute on-site technical support, coordinate with international teams, ensure safety compliance, and deliver reliable engineering solutions that reduce downtime and improve operational continuity. Strong customer communication skills, high adaptability to field environments, and full availability for international travel and on-site service operations.

CORE COMPETENCIES

Field Service & Mechanical Systems: Field Service Engineering • Mechanical Equipment Maintenance • Installation & Commissioning • Industrial Machinery Troubleshooting • Mechanical Systems Validation • Preventive & Corrective Maintenance

Technical: Hydraulics & Fluid Systems • Piping & Mechanical Assemblies • Industrial Equipment • TBM Mechanical Systems • Pressure Testing • Inspection & Verification

Tools: SAP • AutoCAD • Inventor • MS Office • MS Office (advanced Excel, Word, PowerPoint)

PROFESSIONAL EXPERIENCE

Project Engineer / Field Mechanical Engineer | Herrenknecht Tunnelling Services Panama Corp.

Panamá City, Panamá • Jan 2025 – Jul 2025

Global tunnelling OEM supporting international heavy industrial equipment and TBM mechanical systems across multiple projects including USA and Latin America.

- ▶ Developed and standardized a global SOP for critical mechanical maintenance operations on TBM systems covering 3+ active international project sites, improving execution repeatability and reducing non-conformance events.
- ▶ Performed structural validation calculations and functional analysis of mechanical assemblies — including main drive components — ensuring safe operation under high-demand industrial conditions; findings contributed to reducing equipment downtime.
- ▶ Ensured regulatory compliance and documentary traceability across project milestones, improving audit readiness and reducing contractual risk exposure by consolidating a previously fragmented documentation system.
- ▶ Supported engineering reporting and technical coordination between field operations and project stakeholders across USA and international teams.

Key projects: Hudson River Tunnel Project (USA) • Railway Tunnel Project (Perú)

Field Engineer – TBM & Surface Facilities (Intern) | China Railway Tunnel Group Co., Ltd.

Panamá City, Panamá • Aug 2024 – Jan 2025

Contributed to the active tunnelling operations of Panama Metro Line 3, the country's largest infrastructure project.

- ▶ Designed corrective engineering solutions for mechanical and piping systems under unstable terrain conditions, eliminating severe operational incidents and improving continuity of TBM advance.
- ▶ Developed standardized technical documentation (method statements, inspection checklists, work instructions) adopted as reference material by QC and Engineering teams, reducing execution errors by 20%.
- ▶ Produced P&IDs and isometric drawings for high-pressure fluid systems in compliance with ASME B31 standards.
- ▶ Supported pressure testing activities, mechanical assembly verification, and system inspections to ensure reliability and safety of critical TBM subsystems.

Key project: Panama Metro Line 3 – Tunnel Section

EDUCATION

B.Sc. in Mechanical Engineering • Technological University of Panama

Panamá City, Panamá • Jan 2018 – Aug 2024

Relevant coursework: Finite Element Analysis (FEA), Computational Modeling, Numerical Methods, Control Systems, Fluid Mechanics, Thermodynamics.

CERTIFICATIONS & TRAINING

- ▶ Fluid Technology, Hydraulics & Lubrication – Herrenknecht AG (Jun 2025)
- ▶ Design of Thermofluidic Systems – Technological University of Panama (Aug 2023)

FIELD SERVICE AVAILABILITY:

- ▶ Available for international travel (80%+)
- ▶ Valid passport
- ▶ Available for on-call and field service support
- ▶ International project experience
- ▶ Strong adaptability to industrial and offshore environments

LANGUAGES

Spanish (Native) • **English** (C1 – Full Professional Proficiency) • **German** (A2 – actively developing)

PROFESSIONAL AFFILIATIONS

- ▶ Member, Junta Técnica de Ingenieros y Arquitectos de Panamá (JTIA) – (ID: 2025-016-053)